Name: Class: Date:

Activity 1 · Hands-on · 🏠 At home · ⏱ 30 minutes

🥔 Food Tests: Starch and Fat

You will learn: Iodine turns starch blue-black; fats leave a translucent spot on paper.

YOU NEED

- Tincture iodine (from a chemist) diluted with water
- Brown paper or notebook paper
- Foods: potato, rice, bread, sugar, groundnut, coconut, oil, apple
- Small plates, a dropper

⚠️ **Safety:** Do not eat food after adding iodine. Iodine is poisonous if swallowed.

STEPS

1. Put a small piece of each food on a plate.
2. Add one drop of iodine to each and note the colour.
3. Rub each food on the paper and let it dry for 10 minutes.
4. Hold the paper against light — a translucent spot means fat.
5. Record your results.

OBSERVATION TABLE

Food	Colour with iodine	Starch present?	Paper spot	Fat present?

QUESTIONS

1. Which foods had starch? Which had fat?

2. Why does sugar not turn blue-black?

3. In which form do animals store carbohydrate?

WHAT IS HAPPENING? — THE SCIENCE

Starch, a polysaccharide of glucose, forms helices that trap iodine and give a blue-black colour. Cellulose and simple sugars do not. Lipids do not evaporate, so they leave a translucent spot on paper.

Key terms: starch · polysaccharide · lipids · iodine test

Activity 2 · Hands-on · 🧪 In the lab · ⏱ 40 minutes

🫧 Potato Catalase: Enzymes at Work

> **You will learn:** Enzymes speed up reactions and are destroyed (denatured) by high temperature and changed by pH.

YOU NEED

- 3% hydrogen peroxide (from a chemist)
- A raw potato
- 5 small glasses or test tubes
- A grater, a small pan for boiling, ice
- Vinegar and baking soda
- A ruler

> ⚠️ **Safety:** Wear gloves and glasses if available; H_2O_2 can irritate skin and eyes. Do boiling with an adult.

STEPS

1. Grate the potato and make 5 equal portions.
2. Keep one raw, boil one for 5 min, chill one on ice, add a spoon of vinegar to one, and a pinch of baking soda to one.
3. Put each portion in a glass and add the same amount of H_2O_2 (about 2 spoons).
4. After 1 minute, measure the height of the foam with a ruler.
5. Record and compare.

OBSERVATION TABLE

Potato sample	Bubbles (none / few / many)	Foam height (cm)

QUESTIONS

1. Which sample gave the most foam? Why?

2. Why did boiled potato give almost no bubbles?

3. What does an enzyme do to activation energy?

WHAT IS HAPPENING? — THE SCIENCE

Potato cells contain catalase, which breaks hydrogen peroxide into water and oxygen gas (the bubbles). Boiling changes the enzyme's shape (denaturation), so no bubbles. Cold slows it down; acid or alkali changes the active site and reduces activity — showing the effect of temperature and pH.

Key terms: **enzyme** **catalase** **denaturation** **active site** **activation energy**

Digital activity

💻 Enzyme Activity Simulator

> Change the temperature and pH and watch how fast the enzyme works.

Do this digital activity in the BAIO app: **https://narendra-bio-sir.github.io/BAIO-Learning-System-By-Narendra-Sir/activities.html#pu1-ch9/2**

> **You will learn:** Every enzyme has an optimum temperature and an optimum pH. Too hot denatures it permanently; too cold only slows it down.

THINK & WRITE

1. At what temperature and pH was the rate highest?

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2. Why does activity fall sharply above the optimum temperature?
